MADU TEJA SREE

AI Engineer | Generative AI, Machine Learning & MLOps

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building, evaluating and deploying Generative AI and Machine Learning solutions: retrieval-augmented generation (RAG), tool-calling agents, embeddings, deep learning (CNN) and enterprise NLP-to-SQL systems for SAP/ERP data. Applies MLOps practices (containerized model serving, evaluation datasets, regression testing, accuracy and latency validation) to take AI prototypes to production. Strong in data cleaning, validation and quality assurance across large, complex enterprise datasets, and in working directly with clients to turn business requirements into working AI solutions.

EXPERIENCE

AI Engineer – Automatonsx | Nov 2025 – Jun 2026 | Hyderabad, India

Forward Deployed Engineering (FDE) team delivering SAP-integrated AI platforms for enterprise clients (products: Connect and Echo)

- **Data pipeline and quality (Connect):** Built the SAP HANA to PostgreSQL extraction pipeline covering full SAP modules, with schema mapping, data cleaning and validation to ensure accuracy and consistency for downstream AI and analytics. Added configuration-driven switching between two databases through a config file. Fixed an IST/UTC timezone bug that caused timestamp inconsistencies, and diagnosed a missing SAP HANA connection blocking a client pipeline.
- **LLM prompt services and scheduling (Connect):** Built the prompt services for an LLM-driven ETL platform where users describe transformations in plain English and the model returns structured output (regex patterns, join/split configs, PostgreSQL queries) for the data engine to execute, with models served via Ollama/vLLM. Implemented automated pipeline runs with APScheduler (timezone-aware cron jobs), and built the Next.js and React Flow visual flow-builder frontend.
- **Generative AI, NLP-to-SQL (Echo):** Contributed to an enterprise NLP-to-SQL engine on Vanna.ai for SAP/ERP data. Its semantic layer (table relationships, synonyms, metric definitions, value mappings) is injected into every LLM prompt to produce accurate SQL on complex SAP schemas where a generic model would hallucinate.
- **Model evaluation:** Built gold-standard evaluation datasets and semantic mappings across two SAP modules (P2P and FI/CO, 10–11 ECC tables each, e.g. MARA, EKKO, EKPO, BKPF, BSEG) to measure and improve SQL-generation accuracy. Designed two demo tenant datasets from scratch (25-table Indian retail tenant across six domains; Philippine commercial banking tenant), each with a gold dataset and benchmark suite.
- **MLOps and model serving:** Configured and validated self-hosted vLLM endpoints, embedding models and tenant-specific semantic layers; ran structured test queries to validate accuracy, latency and API connectivity. Supported deployment on Docker, PostgreSQL and Linux/Windows VMs, and built a React frontend for the query interface.
- **Productionization:** Independently rebuilt a RAG research agent into a production-ready app with a FastAPI service layer, multi-format ingestion, local embeddings and a tool-calling agent (GPT-4o-mini), plus handoff documentation scoping testing, authentication and deployment.
- **Automation:** Used Claude Code to accelerate development, debugging and documentation, and automated repetitive manual steps across data pipelines and deployment workflows. Earned Azure AI Engineer Associate and Azure AI Fundamentals (Feb 2026) alongside an internal AI bootcamp.

AI Associate Engineer – Datavalley AI | Jun 2025 – Nov 2025 | Hyderabad, India

Project: RAG Research Agent – LangChain, OpenAI, ChromaDB, FastAPI, Docker

- Independently built a tool-calling RAG Research Agent (LangChain + GPT-4o-mini) that routes dynamically between a local knowledge base and live web search based on query intent, rather than a fixed retrieval pipeline.
- Designed a multi-format ingestion pipeline (.txt, .pdf, .docx) with local HuggingFace embeddings and ChromaDB, using content-hash manifests for incremental indexing to avoid full re-ingestion.
- Exposed the agent as a FastAPI service, containerized it with Docker, and built a golden-dataset evaluation harness to regression-test retrieval and answer quality.
- Learned PostgreSQL design and optimization, backend API integration, cloud deployment workflows and collaborative development with Git/GitHub.

PROJECTS

Coffee Sales Revenue Prediction (Machine Learning) | Python, scikit-learn, Pandas, Matplotlib

- Cleaned and prepared transaction data (duplicate and null checks, feature selection, one-hot encoding, scaling) and built scikit-learn pipelines for regression, with exploratory analysis of revenue by coffee type, time of day, weekday and month.
- Compared Linear Regression and Random Forest using MAE, RMSE and R^2 (Random Forest R^2 0.978, MAE 0.27; Linear Regression R^2 0.977, MAE 0.38), with 5-fold cross-validation and GridSearchCV hyperparameter tuning.

Estimation of Nutrition Components from Food Using Deep Learning | Apr 2025 | *Published in IJMTST*

- Designed and trained a CNN-based machine learning model to estimate calories, protein, fat and carbohydrates from food images, with image preprocessing for accurate recognition.
- Trained and validated on real-world food datasets, improving prediction accuracy through feature extraction and optimization.

SKILLS

Generative AI / LLM: RAG, Generative AI, Prompt Engineering, Embeddings, Tool-Calling Agents, AI Agents, Agent Orchestration, Memory Management, Context Engineering, Retrieval Engineering

Machine Learning & Data: Machine Learning, Regression, Random Forest, scikit-learn, Cross-Validation, Hyperparameter Tuning, Deep Learning (CNN), Model Evaluation, Data Cleaning & Validation, Data Quality, NumPy, Pandas, ETL, Gold-Dataset Construction, Semantic Layer Design

MLOps & Model Serving: vLLM, Docker, FastAPI, Evaluation Pipelines, Regression Testing, Deployment Workflows, Git/GitHub

Frameworks: LangChain, LangGraph, LlamaIndex, MCP, Hugging Face, Transformers, CrewAI, Redis, Ollama, APScheduler

Vector DB: ChromaDB, pgvector, FAISS

Backend & Databases: Python, FastAPI, REST APIs, API Integration, PostgreSQL, MySQL, SQL

Cloud, Infra & Web: Azure, Docker, Linux (Ubuntu/RHEL), React, Next.js, React Flow, HTML, CSS

Languages: Python, SQL, C

EDUCATION, CERTIFICATIONS & PUBLICATION

B.Tech – Vijaya Institute of Technology for Women, Vijayawada, AP | Nov 2021 – Apr 2025 | CGPA: 8.13/10.0

Certifications: Microsoft Certified: Azure AI Engineer Associate (Feb 2026); Microsoft Certified: Azure AI Fundamentals (Feb 2026); Introduction to Cybersecurity (CISCO); MongoDB Basics (MongoDB); Python Basic (INFOSYS); Artificial Intelligence (AICTE)

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